Laurent Ortiz

SOC LAB 003

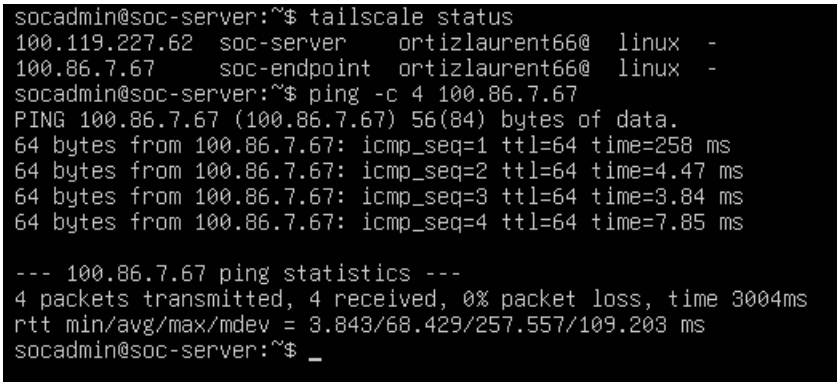
CYBR 4550

10/7/26

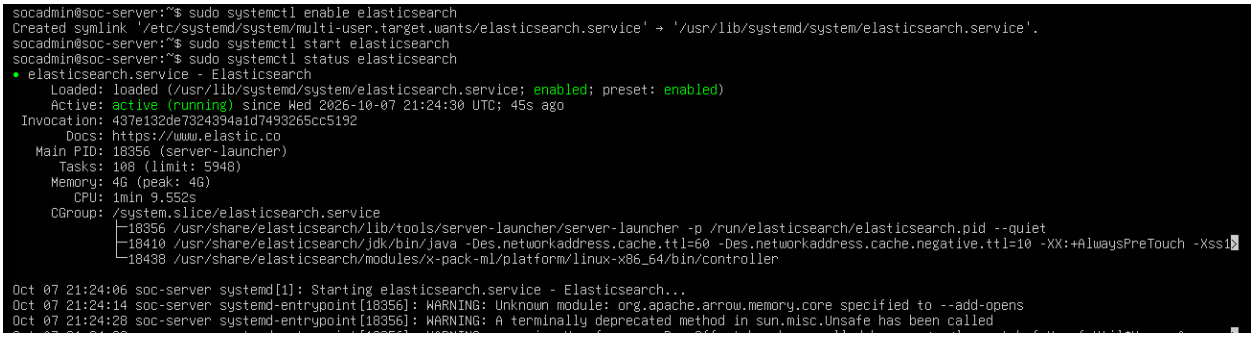
SOC Server Host:



Tailscale Status on both VMs and Ping Endpoint:



Elasticsearch Active:



Elasticsearch listening to port 9200 and response:

```
socadmin@soc-server:~$ sudo ss -lntp | grep 9200
LISTEN 0      4096                         *:9200            *:*    users:(("java",pid=18410,fd=886))
socadmin@soc-server:~$ curl -X GET "localhost:9200"
curl: (52) Empty reply from server
socadmin@soc-server:~$ sudo cat /etc/elasticsearch/elasticsearch.yml | grep xpack.security
xpack.security.enabled: true
xpack.security.enrollment.enabled: true
xpack.security.http.ssl:
xpack.security.transport.ssl:
socadmin@soc-server:~$ sudo journalctl -u elasticsearch -n 50
Oct 07 21:24:06 soc-server systemd[1]: Starting elasticsearch.service - Elasticsearch...
Oct 07 21:24:14 soc-server systemd-entrypoint[18356]: WARNING: Unknown module: org.apache.arrow.memory.core specified to --add-opens
Oct 07 21:24:28 soc-server systemd-entrypoint[18356]: WARNING: A terminally deprecated method in sun.misc.Unsafe has been called
Oct 07 21:24:28 soc-server systemd-entrypoint[18356]: WARNING: sun.misc.Unsafe::arrayBaseOffset has been called by com.google.protobuf.UnsafeUtil$MemoryAccesso>
Oct 07 21:24:28 soc-server systemd-entrypoint[18356]: WARNING: Please consider reporting this to the maintainers of class com.google.protobuf.UnsafeUtil$Memory>
Oct 07 21:24:28 soc-server systemd-entrypoint[18356]: WARNING: sun.misc.Unsafe::arrayBaseOffset will be removed in a future release
Oct 07 21:24:30 soc-server systemd[1]: Started elasticsearch.service - Elasticsearch.

socadmin@soc-server:~$ curl -k https://localhost:9200
{"error":{"root_cause":[{"type":"security_exception","reason":"missing authentication credentials for REST request [/]","header":{"WWW-Authenticate":["Basic rea
lm=\"security\", charset=\"UTF-8\"","Bearer realm=\"security\"","ApiKey"]}}],"type":"security_exception","reason":"missing authentication credentials for REST r
equest [/]","header":{"WWW-Authenticate":["Basic realm=\"security\", charset=\"UTF-8\"","Bearer realm=\"security\"","ApiKey"]}},"status":401}socadmin@soc-server
:~$ ^C
socadmin@soc-server:~$ curl -X GET "localhost:9200"
curl: (52) Empty reply from server
socadmin@soc-server:~$
```

Logstash Active {replaced with docker}:

```
socadmin@soc-server:~$ sudo systemctl status logstash
* logstash.service - logstash
     Loaded: loaded (/usr/lib/systemd/system/logstash.service; disabled; preset: enabled)
     Active: inactive (dead)
socadmin@soc-server:~$ sudo systemctl enable --now logstash
Created symlink '/etc/systemd/system/multi-user.target.wants/logstash.service' → '/usr/lib/systemd/system/logstash.service'.
socadmin@soc-server:~$ sudo systemctl status logstash
● logstash.service - logstash
     Loaded: loaded (/usr/lib/systemd/system/logstash.service; enabled; preset: enabled)
     Active: active (running) since Wed 2026-10-07 21:55:32 UTC; 6s ago
 Invocation: d280b74786414ea6916fd365d7b1c208
   Main PID: 19141 (java)
      Tasks: 24 (limit: 5948)
     Memory: 396.5M (peak: 396.5M)
        CPU: 19.706s
     CGroup: /system.slice/logstash.service
             └─19141 /usr/share/logstash/jdk/bin/java -Xms1g -Xmx1g -Djava.awt.headless=true -Dfile.encoding=UTF-8 -XX:+HeapDumpOnOutOfMemoryError -Djava.secur>

Oct 07 21:55:32 soc-server systemd[1]: Started logstash.service - logstash.
Oct 07 21:55:32 soc-server logstash[19141]: Using bundled JDK: /usr/share/logstash/jdk
lines 1-13/13 (END)
```

Successfully installed Docker with [Elasticsearch + Kibana]:

```
 => [kibana] resolving provenance for metadata file                                                                0.0s
[+] Building 2/2
 ✔ elasticsearch  Built                                                                                           0.0s
 ✔ kibana         Built                                                                                           0.0s
  [ok]   images built: soc-lab/elasticsearch and soc-lab/kibana (9.5.5)

==> 4/6  Starting the stack
[+] Running 4/4
 ✔ Network soc-lab_default      Created                                                                           0.1s
 ✔ Container soc-elasticsearch  Healthy                                                                          48.3s
 ✔ Container soc-setup          Exited                                                                           49.2s
 ✔ Container soc-kibana         Started                                                                          49.5s
  waiting for Kibana to report available..
```

Making endpoint bundle:

```
socanalyst@soc-endpoint:~$ ssh socanalyst@100.86.7.67
The authenticity of host '100.86.7.67 (100.86.7.67)' can't be established.
ED25519 key fingerprint is: SHA256:5FKmkg0ppHDFvOtSpvy2plbVouuvzQf3gTcmJQRqPQ4
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? y
Please type 'yes', 'no' or the fingerprint: yes
Warning: Permanently added '100.86.7.67' (ED25519) to the list of known hosts.
socanalyst@100.86.7.67's password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-34-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Sat Oct 17 12:13:26 AM UTC 2026

  System load:  0.0               Processes:             223
  Usage of /:   39.9% of 13.67GB   Users logged in:       0
  Memory usage: 13%               IPv4 address for ens33: 192.168.80.128
  Swap usage:   0%


Expanded Security Maintenance for Applications is not enabled.

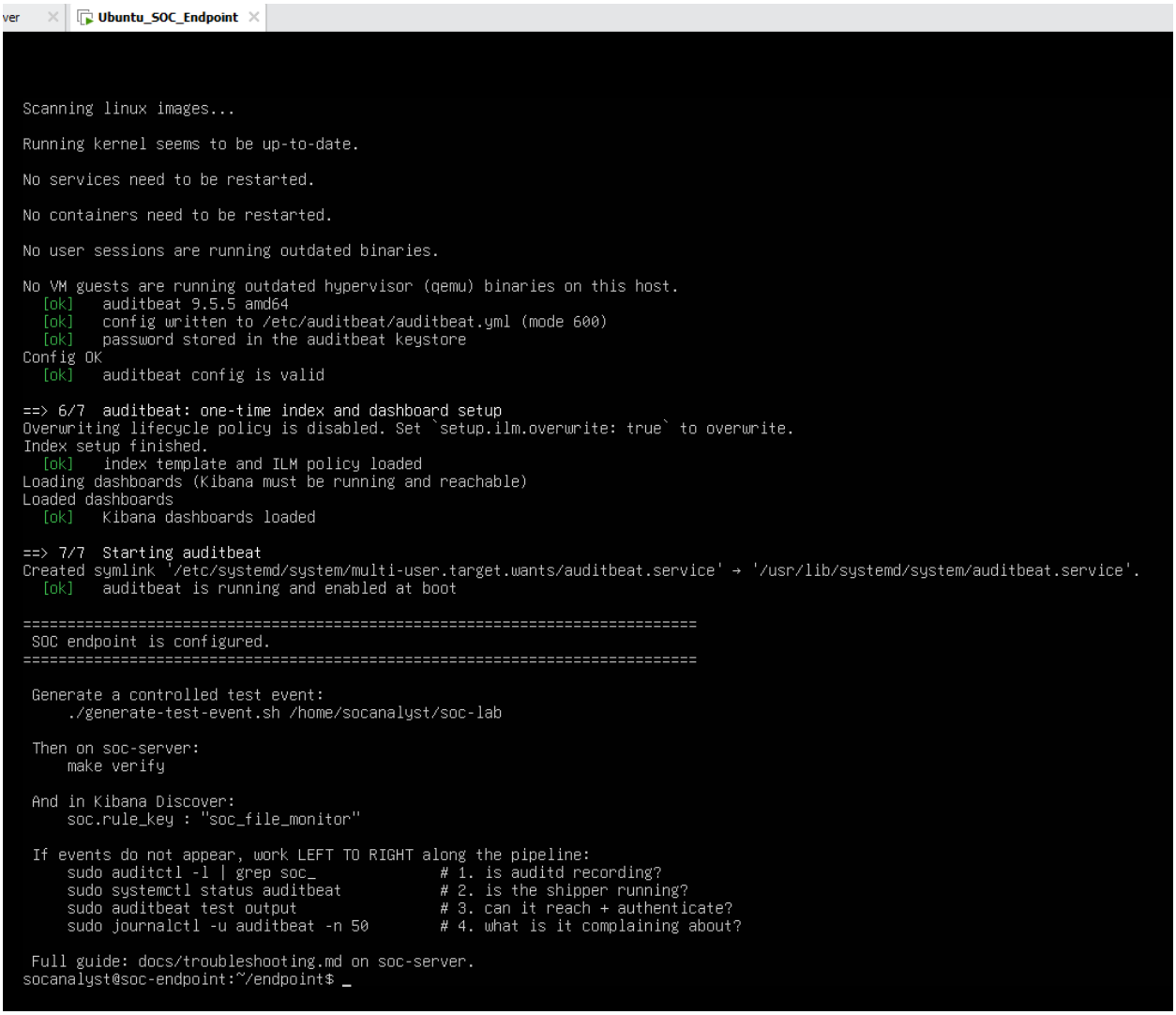
33 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status


The list of available updates is more than a week old.
To check for new updates run: sudo apt update

Last login: Tue Sep 29 02:22:01 2026 from 100.119.227.62
socanalyst@soc-endpoint:~$ ls
auditbeat-9.5.3-linux-x86_64  auditbeat-9.5.3-linux-x86_64.tar.gz  endpoint-setup.tar.gz  soc-lab  soc-lab-info.txt
socanalyst@soc-endpoint:~$
```

Installing bundle package (endpoint-setup.tar.gz) on SOC endpoint:



Endpoint generated test, /home/socanalyst/soc-lab/security-test.txt:

```
==> 2026-10-17T22:28:10Z: writing to /home/socanalyst/soc-lab/security-test.txt
==> local auditd view of the event
----
type=PROCTITLE msg=audit(10/17/2026 22:28:10.732:867) : proctitle=bash ./generate-test-event.sh /home/socanalyst/soc-lab
type=PATH msg=audit(10/17/2026 22:28:10.732:867) : item=0 name=/home/socanalyst/soc-lab/security-test.txt inode=524428 dev=fc:00 mode=file,664 ouid=socanalyst o
gid=socanalyst rdev=00:00 nametype=NORMAL cap_fp=none cap_fi=none cap_fe=0 cap_fver=0 cap_frootid=0
type=CWD msg=audit(10/17/2026 22:28:10.732:867) : cwd=/home/socanalyst/endpoint
type=SYSCALL msg=audit(10/17/2026 22:28:10.732:867) : arch=x86_64 syscall=openat success=yes exit=3 a0=AT_FDCWD a1=0x627907348f20 a2=O_WRONLY|O_CREAT|O_APPEND a
3=0x1b6 items=1 ppid=2500 pid=21433 auid=socanalyst uid=socanalyst gid=socanalyst euid=socanalyst suid=socanalyst fsuid=socanalyst egid=socanalyst sgid=socanaly
st fsgid=socanalyst tty=pts0 ses=5 comm=bash exe=/usr/bin/bash subj=unconfined key=soc_file_monitor

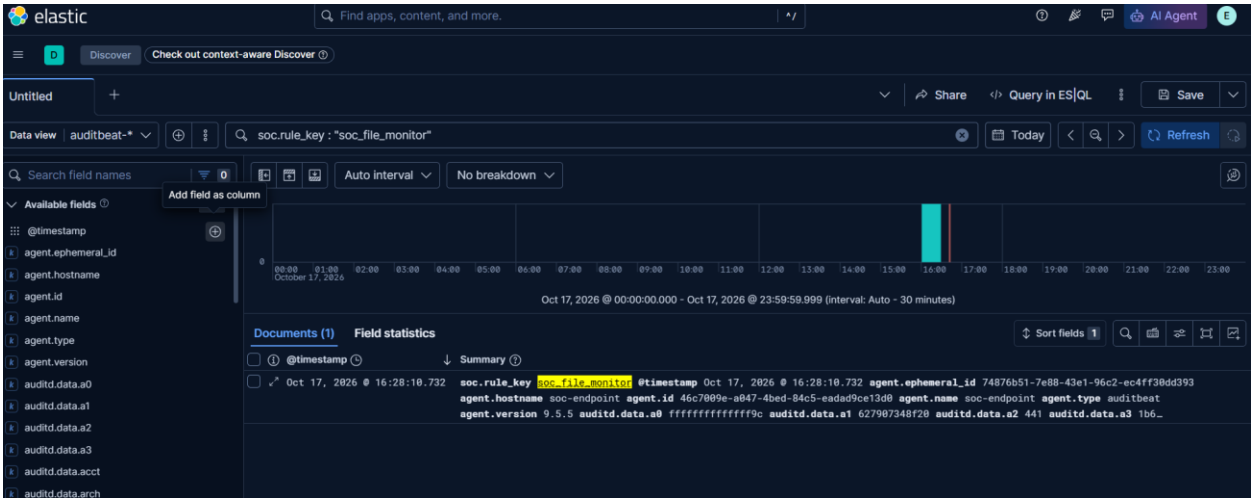
Event generated at: 2026-10-17T22:28:10Z
File:               /home/socanalyst/soc-lab/security-test.txt
User:               socanalyst  (auid: 1000)
Host:               soc-endpoint

Now find the same event centrally, in Kibana Discover:

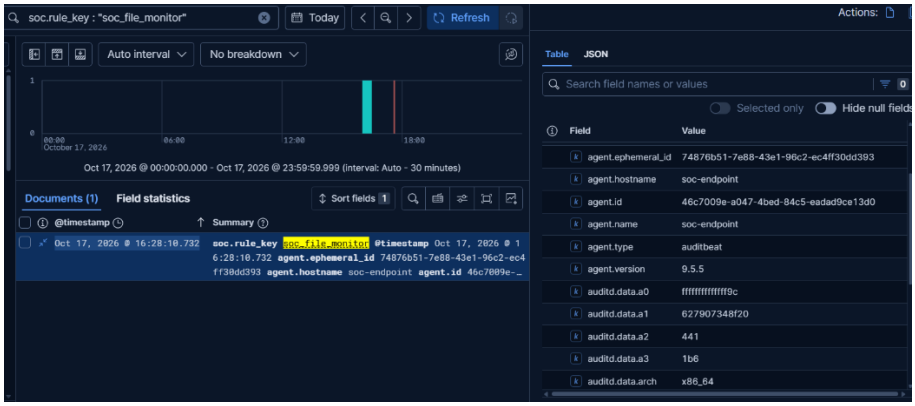
  soc.rule_key : "soc_file_monitor"
  soc.rule_key : "soc_file_monitor" and host.name : "soc-endpoint"

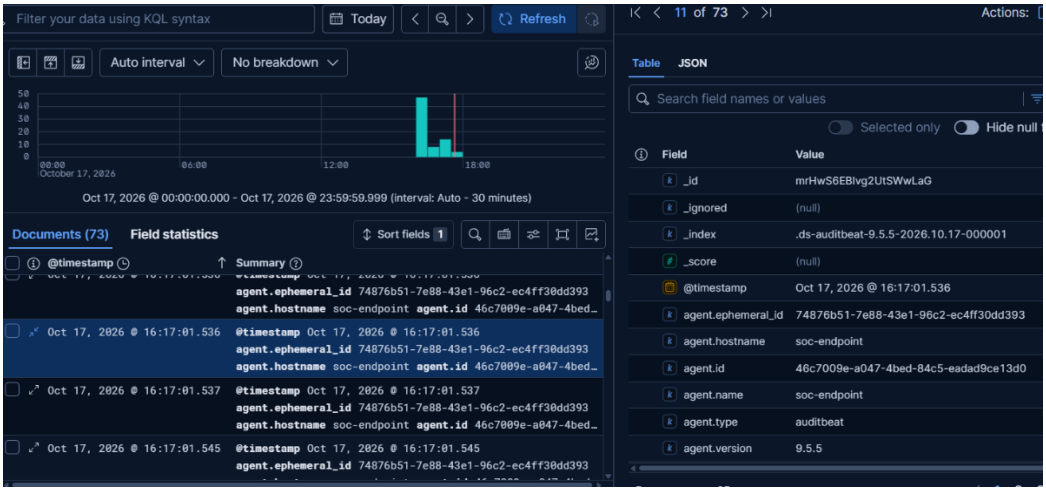
Record the timestamp above - you need it for the investigation timeline.
socanalyst@soc-endpoint:~/endpoint$ sudo tail -n 30 /var/log/auditbeat/auditbeat
tail: cannot open '/var/log/auditbeat/auditbeat' for reading: No such file or directory
socanalyst@soc-endpoint:~/endpoint$ sudo ls -lh /var/log/auditbeat/
total 100K
-rw-r----- 1 root root 9.5K Oct 17 00:56 auditbeat-20261017-12.ndjson
-rw-r----- 1 root root 7.0K Oct 17 00:57 auditbeat-20261017-13.ndjson
-rw-r----- 1 root root 1.1K Oct 17 00:58 auditbeat-20261017-14.ndjson
-rw-r----- 1 root root  702 Oct 17 21:45 auditbeat-20261017-15.ndjson
-rw-r----- 1 root root  702 Oct 17 22:11 auditbeat-20261017-16.ndjson
-rw-r----- 1 root root  702 Oct 17 22:13 auditbeat-20261017-17.ndjson
-rw-r----- 1 root root  56K Oct 17 22:29 auditbeat-20261017-18.ndjson
-rw-r----- 1 root root 1.5K Oct 17 22:15 auditbeat-20261017-19.ndjson
socanalyst@soc-endpoint:~/endpoint$
```

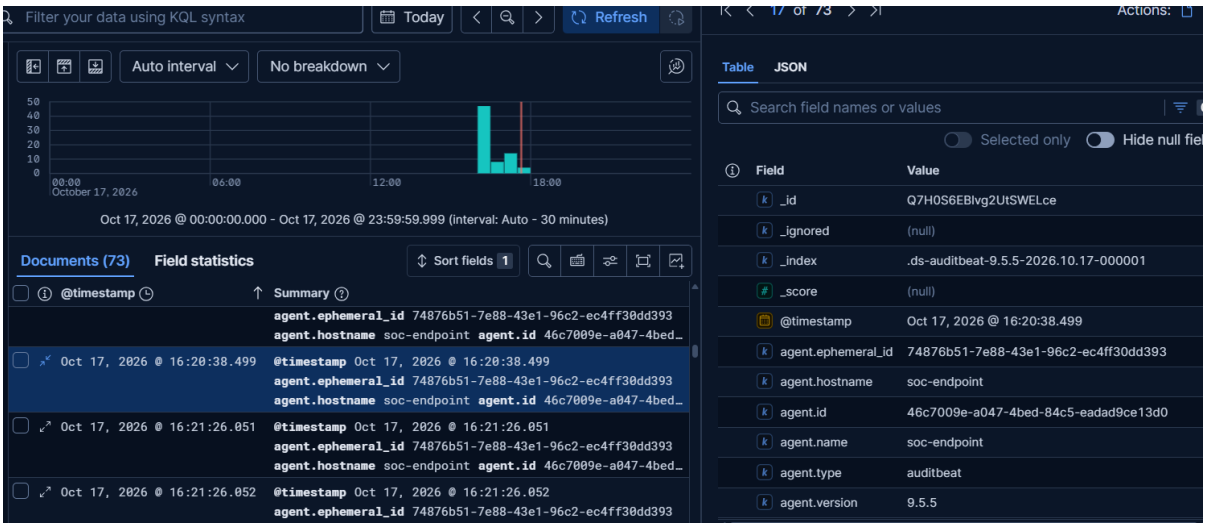
Elastic Kibana, showing soc_file_monitor rule working:

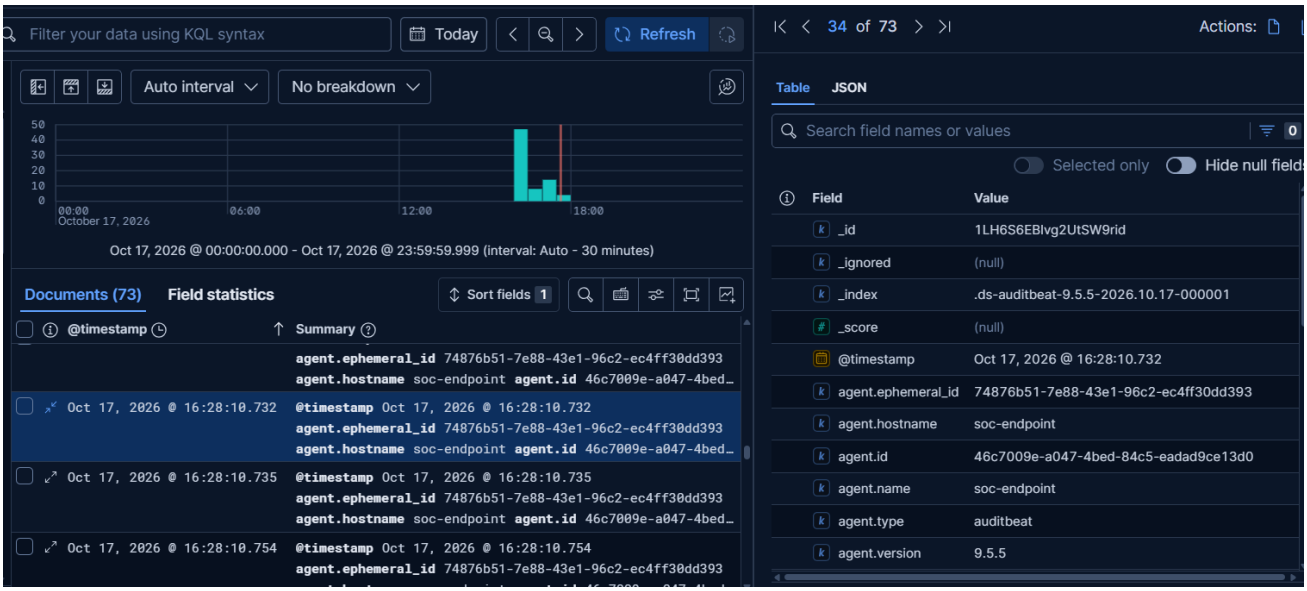


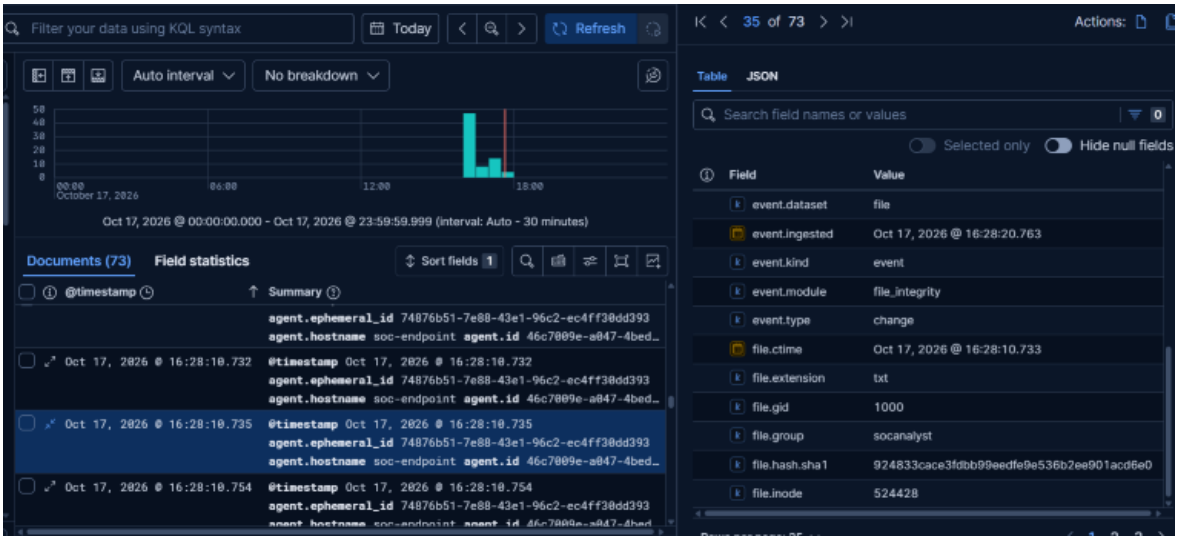
Screenshots for timeline evidence:

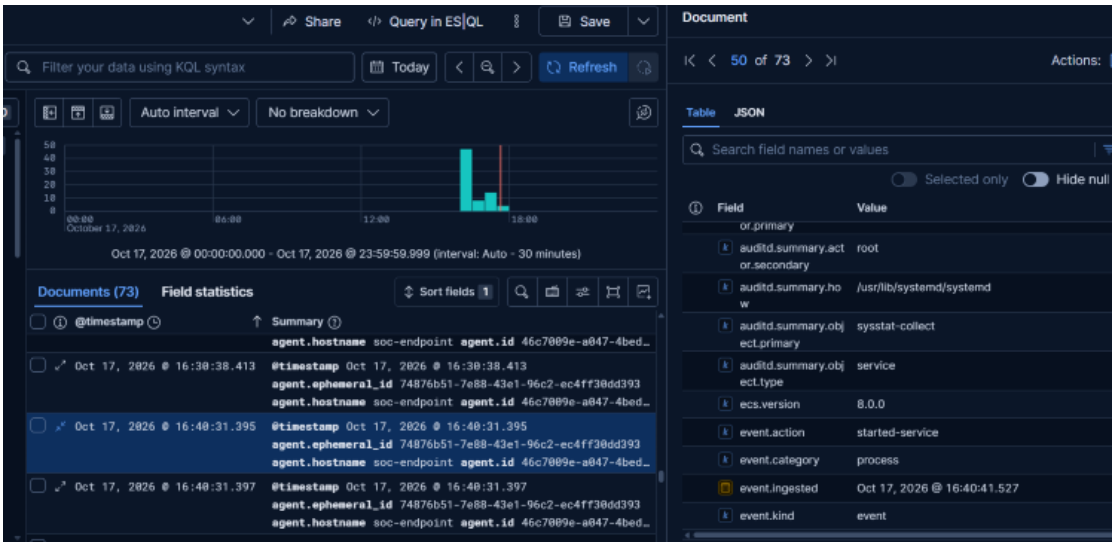


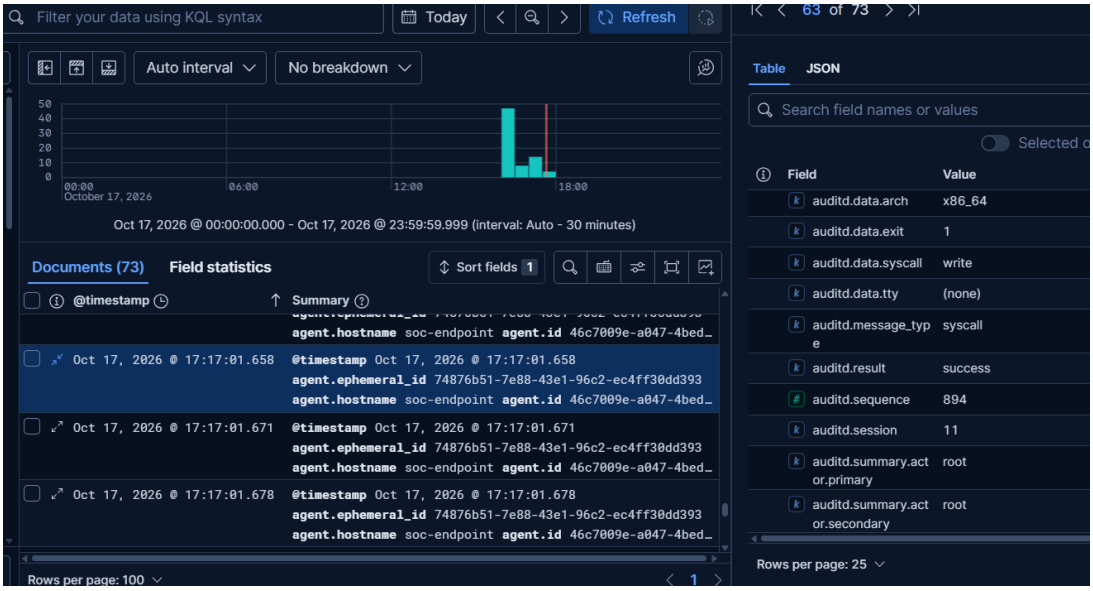

Filter your data using KQL syntax
Today
Refresh
Auto interval
No breakdown
Oct 17, 2026 @ 00:00:00.000 - Oct 17, 2026 @ 23:59:59.999 (interval: Auto - 30 minutes)
Documents (73)
Field statistics
Sort fields 1
@timestamp
Summary
Oct 17, 2026 @ 16:17:01.536
Oct 17, 2026 @ 16:17:01.537
Oct 17, 2026 @ 16:17:01.545
11 of 73
Actions:
Table
JSON
Search field names or values
Selected only
Hide null
Field
Value
_id
mrHwS6EBlvg2UtSWwLaG
_ignored
(null)
_index
.ds-auditbeat-9.5.5-2026.10.17-000001
_score
(null)
@timestamp
Oct 17, 2026 @ 16:17:01.536
agent.ephemeral_id
74876b51-7e88-43e1-96c2-ec4ff30dd393
agent.hostname
soc-endpoint
agent.id
46c7009e-a047-4bed-84c5-eadad9ce13d0
agent.name
soc-endpoint
agent.type
auditbeat
agent.version
9.5.5


Filter your data using KQL syntax
Today
Refresh
Auto interval
No breakdown
Oct 17, 2026 @ 00:00:00.000 - Oct 17, 2026 @ 23:59:59.999 (interval: Auto - 30 minutes)
Documents (73)
Field statistics
Sort fields 1
@timestamp
Summary
Oct 17, 2026 @ 16:20:38.499
Oct 17, 2026 @ 16:21:26.051
Oct 17, 2026 @ 16:21:26.052
17 of 73
Actions:
Table
JSON
Search field names or values
Selected only
Hide null fie
Field
Value
_id
Q7H0S6EBlvg2UtSWELce
_ignored
(null)
_index
.ds-auditbeat-9.5.5-2026.10.17-000001
_score
(null)
@timestamp
Oct 17, 2026 @ 16:20:38.499
agent.ephemeral_id
74876b51-7e88-43e1-96c2-ec4ff30dd393
agent.hostname
soc-endpoint
agent.id
46c7009e-a047-4bed-84c5-eadad9ce13d0
agent.name
soc-endpoint
agent.type
auditbeat
agent.version
9.5.5


Filter your data using KQL syntax
Today
Refresh
Auto interval
No breakdown
Oct 17, 2026 @ 00:00:00.000 - Oct 17, 2026 @ 23:59:59.999 (interval: Auto - 30 minutes)
Documents (73)
Field statistics
Sort fields 1
@timestamp
Summary
Oct 17, 2026 @ 16:28:10.732
Oct 17, 2026 @ 16:28:10.735
Oct 17, 2026 @ 16:28:10.754
34 of 73
Actions:
Table
JSON
Search field names or values
Selected only
Hide null fields
Field
Value
_id
1LH6S6EBlvg2UtSW9rid
_ignored
(null)
_index
.ds-auditbeat-9.5.5-2026.10.17-000001
_score
(null)
@timestamp
Oct 17, 2026 @ 16:28:10.732
agent.ephemeral_id
74876b51-7e88-43e1-96c2-ec4ff30dd393
agent.hostname
soc-endpoint
agent.id
46c7009e-a047-4bed-84c5-eadad9ce13d0
agent.name
soc-endpoint
agent.type
auditbeat
agent.version
9.5.5


Filter your data using KQL syntax
Today
Refresh
Auto interval
No breakdown
Oct 17, 2026 @ 00:00:00.000 - Oct 17, 2026 @ 23:59:59.999 (interval: Auto - 30 minutes)
Documents (73)
Field statistics
Sort fields 1
@timestamp
Summary
35 of 73
Actions:
Table
JSON
Search field names or values
Selected only
Hide null fields
Field
Value
event.dataset file
event.ingested Oct 17, 2026 @ 16:28:20.763
event.kind event
event.module file_integrity
event.type change
file.ctime Oct 17, 2026 @ 16:28:10.733
file.extension txt
file.gid 1000
file.group socanalyst
file.hash.sha1 924833cace3fdbb99eedfe9e536b2ee901acd6e0
file.inode 524428


Share
Query in ES|QL
Save
Filter your data using KQL syntax
Today
Refresh
Auto interval
No breakdown
Oct 17, 2026 @ 00:00:00.000 - Oct 17, 2026 @ 23:59:59.999 (interval: Auto - 30 minutes)
Documents (73)
Field statistics
Sort fields 1
@timestamp
Summary
Document
50 of 73
Actions:
Table
JSON
Search field names or values
Field
Value
auditd.summary.actor.secondary root
auditd.summary.how /usr/lib/systemd/systemd
auditd.summary.object.primary sysstat-collect
auditd.summary.object.type service
ecs.version 8.0.0
event.action started-service
event.category process
event.ingested Oct 17, 2026 @ 16:40:41.527
event.kind event


Filter your data using KQL syntax
Today
Refresh
Auto interval
No breakdown
Oct 17, 2026 @ 00:00:00.000 - Oct 17, 2026 @ 23:59:59.999 (interval: Auto - 30 minutes)
Documents (73)
Field statistics
Sort fields 1
@timestamp
Summary
63 of 73
Table
JSON
Search field names or values
Field
Value
auditd.data.arch x86_64
auditd.data.exit 1
auditd.data.syscall write
auditd.data.tty (none)
auditd.message_type syscall
auditd.result success
auditd.sequence 894
auditd.session 11
auditd.summary.actor.primary root
auditd.summary.actor.secondary root
Rows per page: 100
Rows per page: 25

Investigation Report:

Incident: File Monitoring and Endpoint Activity Investigation

Date: October 17, 2026

Investigator: Laurent Ortiz

Affected Host: soc-endpoint

Final Decision: Benign Activity True Positive Detection

Alert Severity: Medium 3/5

Executive Summary

During my SOC investigation, I analyzed security events collected from one of the Ubuntu virtual machines, soc-endpoint, using Auditd, Auditbeat, Elasticsearch, and Kibana. I reviewed 73 Auditbeat documents and chose six events to investigate system activity after creating a controlled file-monitoring test. The main detection happened when I executed a script that created security-test.txt in a directory. The activity triggered the soc_file_monitor audit rule, and I was able to locate the event in Kibana's discover. My investigation determined that the file creation was benign because it was an authorized test intended to verify the logging pipeline. Even though it's benign, I still classified the detection as a true positive because the monitoring system correctly identified and recorded the activity.

Environment

My SOC environment consisted of two Ubuntu virtual machines connected through Tailscale. The SOC Server (soc-server) was managed through the socadmin account that hosted Elasticsearch and Kibana using Docker containers provided by Professor Haro. Elasticsearch was connected through 100.119.227.62:9200, while Kibana was connected through port 5601. The SOC server was responsible for receiving, storing, and providing centralized access to security events. The SOC Endpoint (soc-endpoint) was managed through the socanalyst account and the monitored system. I configured Auditd to record Linux system activity and Auditbeat 9.5.5 to forward security telemetry to Elasticsearch.

Detection

The primary detection happened when I executed *./generate-test-event.sh /home/socanalyst/soc-lab* on the SOC Endpoint. The script generated file activity involving */home/socanalyst/soc-lab/security-test.txt*. Auditd recorded the openat system call, the

associated user, and the results of the script using the soc_file_monitor audit rule. Auditbeat collected the event and forwarded it to Elasticsearch, and I was able to view the event in Kibana. In Kibana Discover, under the auditbeat-* data view and I searched for: soc.rule_key : "soc_file_monitor". The query returned one matching document from soc-endpoint at 16:28:10.732. There was also another separate file integrity change event that was recorded at 16:28:10.735. These findings verified that the SOC endpoint was generating and forwarding security telemetry.

Investigation Timeline:

Time	Event	System/User	Analyst Observation
Before testing	SOC connectivity established	SOC server - socadmin SOC Endpoint – socanalyst	Tailscale installed to both VMS and the host.
Before testing	Central logging services configured	SOC Server - socadmin	Elasticsearch and Kibana deployed.
Oct 17, 2026 @ 16:17:01.536	Auditd activity recorded	SOC Endpoint – socanalyst	Auditbeat collected a system event before the controlled test.
Oct 17, 2026 @ 16:20:38.499	Additional audit activity	SOC Endpoint - analyst	Another event was indexed before the file-monitoring test. Its specific action would require reviewing more fields.
Oct 17, 2026 @ 16:28:10.732	File-monitoring rule triggered	SOC Endpoint - analyst	The event matches soc_file_monitor and corresponds to the authorized creation of security-test.txt.
Oct 17, 2026 @ 16:28:10.735	File integrity change recorded	SOC Endpoint File owner/group: socanalyst	Auditbeat's file integrity module recorded a file change shortly after the audit rule event.
Oct 17, 2026 @ 16:40:31.395	System service started	SOC Endpoint - System/root	An Auditbeat document records event.action: started-service, involving sysstat-collect.
Oct 17, 2026 @ 17:17:01.658	Successful system call recorded	SOC Endpoint - Root	Auditd recorded a successful write system call, with audit sequence 894.

Technical Analysis

I started my investigation by examining the local Auditd records that were generated on the SOC Endpoint. The main event happened at arount 16:28:10 when I executed the test script. The audit record identified the user as socanalyst (UID 1000), the affected file as */home/socanalyst/soc-lab/security-test.txt*, and the system call as openat. The record showed "success=yes" and "exit=3", meaning that the operation completed successfully and returned file descriptor 3. Auditbeat was configured to collect audit events through its auditd module. I verified that Auditbeat was running as a systemd service and that its connection to Elasticsearch was successful. The SOC endpoint also used the file_integrity module to collect file activity and metadata. During the configuration steps, I kept receiving repeated 401 unauthorized errors because the password configured for the soc-beats account didn't match the credentials stored in Elasticsearch. I troubleshot the issue by verifying the Elasticsearch user, resetting the account password, updating the Auditbeat keystore, and restarting Auditbeat. After resetting the account, I used the command *sudo auditbeat test* and was able to confirm successful authentication. In Elasticsearch, I verified that Auditbeat events were indexed in the auditbeat-* data view. I also confirmed that the soc-lab-normalize ingest pipeline was installed.

Using Kibana Discover from my host, I reviewed 73 Auditbeat documents and chose six events for further investigation. Two events occurred before the controlled test, which were then followed by the file-monitoring detection at 16:28:10.732 and a file integrity change at 16:28:10.735. The file integrity document also showed an ingestion timestamp of 16:28:20.763, so about 10 seconds after the event had occurred. I also examined a service-start event at 16:40:31.395, and a successful write system call at 17:17:01.658. These events collected by Auditbeat are broader endpoint activity. After investigating these events, I didn't find enough evidence to associate these events with malicious behavior.

Alert Severity and Final Determination

I classified the primary event as medium because the file creation was intentionally performed as part of the lab, but in a production environment, it would be considered a more serious event. The action was executed by the expected user (socanalyst) and occurred within the test directory. Although the monitoring rule correctly detected the activity, there was no indication that the file creation involved unauthorized access or malicious intent.

My final determination was Benign Activity (True Positive). The soc_file_monitor rule successfully detected the authorized file activity, and I confirmed that the event was collected through Auditbeat, indexed in Elasticsearch, and visible in Kibana Discover. The

additional system events provided more useful context about endpoint activity, but the evidence didn't establish that they were malicious or related to the primary detection. Therefore, I would document the controlled test as successful and close the primary event as a benign true positive.

Recommended Actions

Based on my investigation of the six events, I would document the detection and close the primary event after confirming that it was an authorized test. I would suggest continued monitoring of the endpoint for any more unexpected file changes, unusual processes, and suspicious user activity. I would also improve the environment by enabling TLS encryption between Auditbeat and Elasticsearch, improving credential management, and configuring automated alerts for higher-risk activity. Because I was experiencing authentication issues during setup, I would also verify that service accounts and keystore credentials remain synchronized. For the system events at 16:17:01 and 16:20:38, I would further investigate to identify their actions and users. I would review the service-start and root-associated system-call events to determine whether they were consistent with normal system operations. These improvements would strengthen monitoring reliability and make the lab more representative of a production SOC environment.

Final Reflection

After completing SOC Labs parts 1 – 3, I learned that SOC operations involve more than just collecting logs. While log collection records system activity, security monitoring involves analyzing those logs to identify potential threats. Centralized logging allows soc analysts to investigate events from multiple endpoints in one location. Kibana Discover made my investigation easier by allowing me to search, filter, and correlate events instead of manually reviewing logs in the terminal at /var/log/audit/audit.log.

My biggest challenge was configuring the logging pipeline between my SOC Endpoint and SOC Server, which took me a few hours to resolve. I kept receiving repeated 401 Unauthorized errors that were caused by mismatched Elasticsearch credentials and issues with the end-to-end diagnostic. I utilized AI to assist me with troubleshooting these problems, and it taught me the importance of verifying each component individually and being patient. I plan to implement these VMs in my home lab, and to make the environment more realistic, I would implement TLS encryption, automated alerting, and add more monitored endpoints. Overall, this lab improved my understanding of how SOC analysts investigate events and distinguish benign activity from potential threats.